

Лабораторная работа 2 ИГИ: Работа с Docker.

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1. Подготовьте рабочее окружение в соответствии с типом вашей операционной системы
 - Установите Docker
 - Выполните базовую настройку
2. Изучите простейшие консольные команды и возможности Docker Desktop (см. лекцию), создать собственный контейнер docker/getting-started, открыть в браузере и изучить tutorial.

```
→ docker run -d -p 80:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
cb9626c74200: Download complete
f1d1c9928c82: Download complete
ee68d3549ec8: Download complete
4f7e34c2de10: Download complete
33e0cbbb4673: Download complete
1e35f6679fab: Download complete
9b6f639ec6ea: Download complete
b6334b6ace34: Download complete
c158987b0551: Download complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
83838c24db6088c150ae747111ff062c8a790edfb8890295a60fb2f59b831f4b
→ LR2 ⚡ (p igi_lr2)
```

Containers

Images

Volumes

Builds

Docker Scout

Extensions

Containers [Give feedback](#)

Container CPU usage

0.00% / 1600% (16 CPUs available)

Container memory usage

12MB / 7.4GB

[Show charts](#)

Only show running containers

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	unruffled_franklin	83838c24db60	docker/getting-started:<none>	80:80	0%	4 minutes ago	
☐	hakaton	-	-	-	0%	1 hour ago	
☐	react_app	1f909a3886c3	hakaton-frontend:<none>	3000:3000	0%	1 hour ago	
☐	proxy_server	82ca6f4ef48a	hakaton-proxy:<none>	8080:8080	0%	1 hour ago	
☐	django_backend	ea6ca449d808	hakaton-django:<none>	8000:8000..	0%	1 hour ago	
☐	redis_cache	b4960a52c07e	redis:alpine	6379:6379	0%	1 hour ago	

Showing 6 items

Getting Started

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Getting Started

Getting Started

Our Application

Updating our App

Sharing our App

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Using Bind Mounts

Multi-Container Apps

Using Docker Compose

Image Building Best Practices

What Next?

Getting Started

The command you just ran

Congratulations! You have started the container for this tutorial! Let's first explain the command that you just ran. In case you forgot, here's the command:

```
docker run -d -p 80:80 docker/getting-started
```

You'll notice a few flags being used. Here's some more info on them:

- `-d` - run the container in detached mode (in the background)
- `-p 80:80` - map port 80 of the host to port 80 in the container
- `docker/getting-started` - the image to use

Pro tip

You can combine single character flags to shorten the full command. As an example, the command above could be written as:

```
docker run -dp 80:80 docker/getting-started
```

Table of contents

The command you just ran

The Docker Dashboard

What is a container?

What is a container image?

3. Создайте docker image, который запускает скрипт с использованием функций из https://github.com/smartikaorg/geometric_lib.

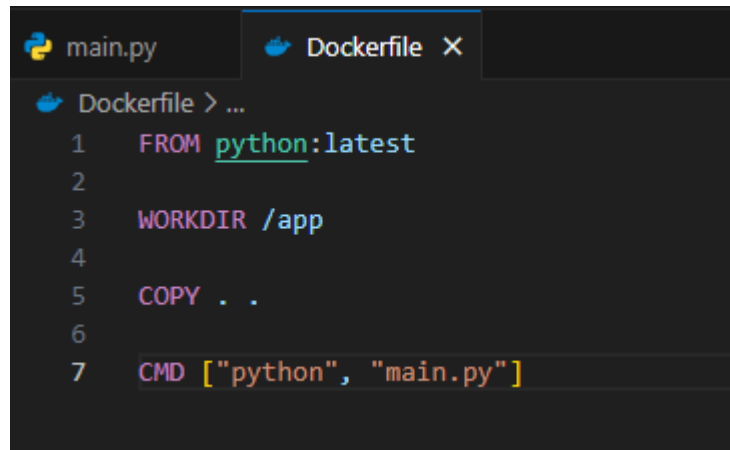
а. Данные необходимые для работы скрипта передайте любым удобным способом (например: конфиг файл через docker volume, переменные окружения, перенаправление ввода). Изучите простейшие консольные команды для работы с docker(см. лекцию). Зарегистрируйтесь на DockerHub и выберите необходимые для проекта образы

```
main.py X
main.py > ...
1 import circle
2 import square
3
4 var = int(input("Enter length of square side: "))
5 print("Square area = ", square.area(var))
6 print("Square perimeter = ", square.perimeter(var))
7 var = int(input("Enter length of circle radius: "))
8 print("Circle area = ", circle.area(var))
9 print("Circle perimeter = ", circle.perimeter(var))

docker pull python
Using default tag: latest
latest: Pulling from library/python
21754c21aa78: Download complete
8031108f3cda: Download complete
854e2aed8deb: Download complete
447713e77b4f: Download complete
155ad54a8b28: Download complete
1d281e50d3e4: Download complete
5e9ad5aa09b4: Download complete
Digest: sha256:385ccb8304f6330738a6d9e6fa0bd7608e006da7e15bc52b33b0398e1ba4a15b
Status: Downloaded newer image for python:latest
docker.io/library/python:latest

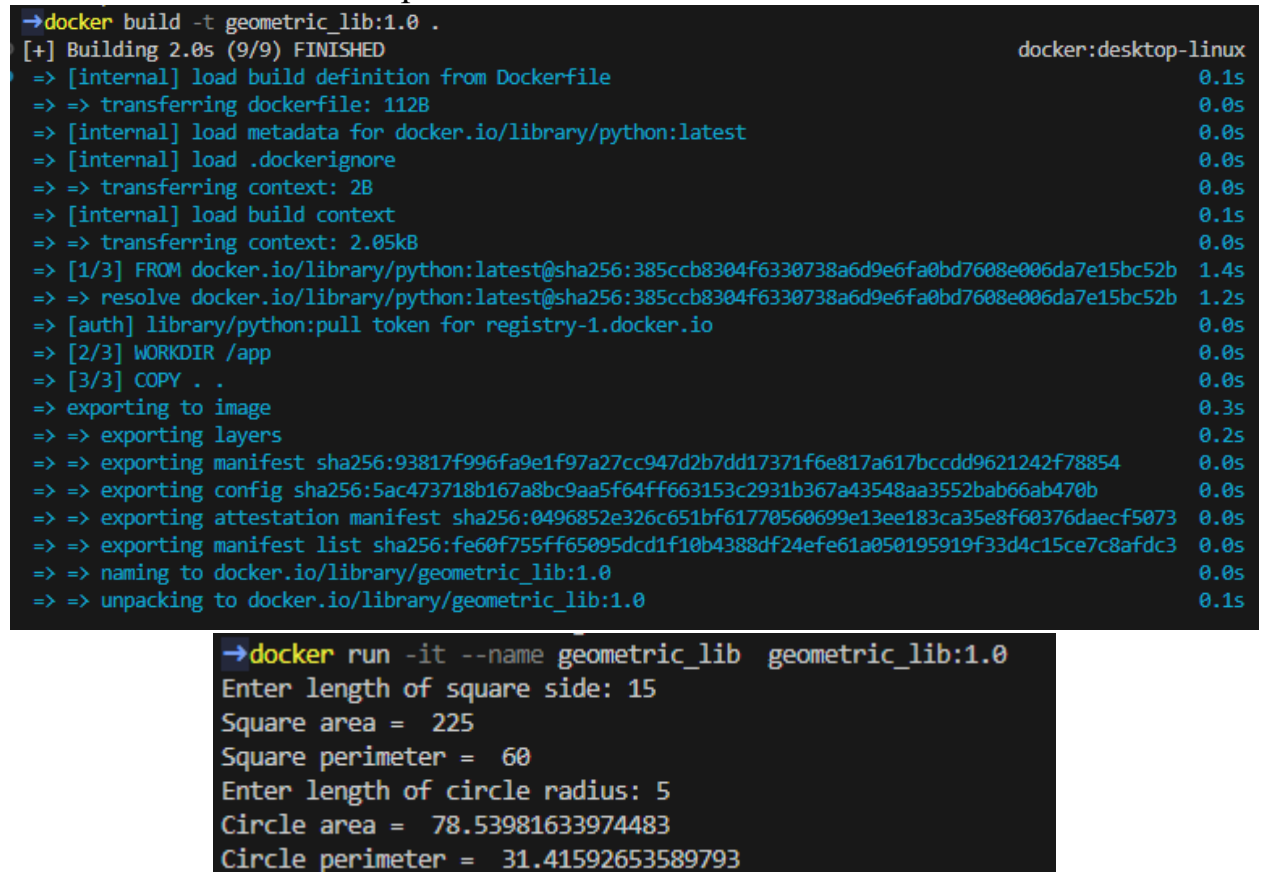
What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview python
```

- б. Создать Dockerfile для реализации сборки собственных Docker образов



```
main.py Dockerfile X
Dockerfile > ...
1 FROM python:latest
2
3 WORKDIR /app
4
5 COPY . .
6
7 CMD ["python", "main.py"]
```

- с. Использовать его для создания контейнера. Протестировать использование контейнера



```
→ docker build -t geometric_lib:1.0 .
[+] Building 2.0s (9/9) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 112B                               0.0s
=> [internal] load metadata for docker.io/library/python:latest 0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                     0.0s
=> [internal] load build context                                0.1s
=> => transferring context: 2.05kB                                0.0s
=> [1/3] FROM docker.io/library/python:latest@sha256:385ccb8304f6330738a6d9e6fa0bd7608e006da7e15bc52b 1.4s
=> => resolve docker.io/library/python:latest@sha256:385ccb8304f6330738a6d9e6fa0bd7608e006da7e15bc52b 1.2s
=> [auth] library/python:pull token for registry-1.docker.io    0.0s
=> [2/3] WORKDIR /app                                           0.0s
=> [3/3] COPY . .                                              0.0s
=> exporting to image                                           0.3s
=> => exporting manifest sha256:93817f996fa9e1f97a27cc947d2b7dd17371f6e817a617bccdd9621242f78854 0.0s
=> => exporting config sha256:5ac473718b167a8bc9aa5f64ff663153c2931b367a43548aa3552bab66ab470b 0.0s
=> => exporting attestation manifest sha256:0496852e326c651bf61770560699e13ee183ca35e8f60376daecf5073 0.0s
=> => exporting manifest list sha256:fe60f755ff65095dcd1f10b4388df24efe61a050195919f33d4c15ce7c8afdc3 0.0s
=> => naming to docker.io/library/geometric_lib:1.0             0.0s
=> => unpacking to docker.io/library/geometric_lib:1.0          0.1s

→ docker run -it --name geometric_lib geometric_lib:1.0
Enter length of square side: 15
Square area = 225
Square perimeter = 60
Enter length of circle radius: 5
Circle area = 78.53981633974483
Circle perimeter = 31.41592653589793
```

4. Скачать любой доступный проект с GitHub с произвольным стеком технологий (пример – см. индивидуальное задание) или использовать свой, ранее разработанный. Создать для него необходимый контейнер, используя Docker Compose для управления многоконтейнерными приложениями. Запустить проект в контейнере.(Примеры Images:

https://hub.docker.com/_/phpmyadmin, https://hub.docker.com/_/mysql,
https://hub.docker.com/_/postgres)

Dockerfile.django:

```
FROM python:3.11-slim

WORKDIR /app

COPY ../ .

RUN pip install -r requirements.txt

RUN python manage.py migrate

EXPOSE 8000 8001

CMD ["sh", "-c", "python manage.py runserver 0.0.0.0:8000 & daphne -b 0.0.0.0 -p 8001 filtering_service.asgi:application"]
```

Dockerfile.proxy:

```
FROM python:3.11-slim

WORKDIR /app

COPY ./proxy_server /app

RUN pip install -r requirements.txt

EXPOSE 8080

CMD ["python", "proxy.py"]
```

Dockerfile:

```
FROM node:16

WORKDIR /app

COPY package*.json ./

RUN npm install

COPY . .

EXPOSE 3000

CMD ["npm", "start"]
```

Docker-compose.yaml:

```
version: '3.8'

services:
  django:
    container_name: django_backend
    build:
      context: ./BACK/filtered_service/filtering_service
      dockerfile: Dockerfile.django
    command: ["sh", "-c", "python manage.py migrate && python manage.py runserver 0.0.0.0:8000 & daphne -b 0.0.0.0 -p 8001 filtering_service.asgi:application"]
    volumes:
      - ./BACK/filtered_service/filtering_service:/app
    ports:
      - '8000:8000'
      - '8001:8001'
    depends_on:
      - redis
    environment:
      - REDIS_HOST=redis
      - REDIS_PORT=6379
    networks:
      - app_network

  proxy:
    container_name: proxy_server
    build:
      context: ./BACK/filtered_service/filtering_service
      dockerfile: Dockerfile.proxy
    volumes:
      - ./BACK/filtered_service/filtering_service/proxy_server:/app
    ports:
      - "8080:8080"
    depends_on:
      - django
    networks:
      - app_network

  redis:
    container_name: redis_cache
    image: 'redis:alpine'
    ports:
      - '6379:6379'
    networks:
      - app_network

  frontend:
    container_name: react_app
    build:
      context: ./FRONT/admin-panel
      dockerfile: Dockerfile
    ports:
```

```

- "3000:3000"

volumes:
- ./FRONT/admin-panel:/app
- /app/node_modules

environment:
- NODE_ENV=development

depends_on:
- django

command: npm start

networks:
- app_network

```

```

networks:
  app_network:
    driver: bridge

```

```

→ docker-compose build
time="2025-03-02T19:08:30+03:00" level=warning msg="D:\BSUIR\4th_Semester\IGI\353502_KOZELLO_13\IGI\LR2\HAKATON\docker-compose.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Building 131.2s (12/12)
[+] Building 222.6s (31/31) FINISHED
    => [django internal] load build definition from Dockerfile.django
    => transferring dockerfile: 314B
    => [proxy internal] load metadata for docker.io/library/python:3.11-slim
    => [django auth] library/python:pull token for registry-1.docker.io
    => [django internal] load .dockerignore
    => transferring context: 2B
    => [proxy 1/4] FROM docker.io/library/python:3.11-slim@sha256:614c8691ab74150465ec9123378cdAdde7a6e57be9e558c3108df40664667a4c
    => resolve docker.io/library/python:3.11-slim@sha256:614c8691ab74150465ec9123378cdAdde7a6e57be9e558c3108df40664667a4c
    => [django internal] load build context
    => transferring context: 159.92MB
    => CACHED [proxy 2/4] WORKDIR /app
    => [django 3/5] COPY ./ .
    => [django 4/5] RUN pip install -r requirements.txt
    => [django 5/5] RUN python manage.py migrate
    => [django] exporting to image
    => exporting layers
    => exporting manifest sha256:ca555d1e4f424ec6cb53d7de8804e9d0fe316696ad9641fe5320854265c7ed6a
    => exporting config sha256:7c4a8d435758606f6558f2c9f8ff59a77c2f5e5b7f5e425f78208764987f76
    => exporting attestation manifest sha256:c07a5c7b55a01cef945e4c09f2104d20b0d15e52a0b7112f013d4547188d36c
    => exporting manifest list sha256:70c1b36e31f54c8bb0333b62c42b4de2201f6461d1d43ff5761e02ba6b357fd
    => naming to docker.io/library/hakaton-django:latest
    => unpacking to docker.io/library/hakaton-django:latest
    => [django] resolving provenance for metadata file
    => [frontend internal] load build definition from Dockerfile
    => transferring dockerfile: 148B
    => [proxy internal] load build definition from Dockerfile.proxy
    => transferring dockerfile: 194B
    => [frontend internal] load metadata for docker.io/library/node:16
    => [frontend auth] library/node:pull token for registry-1.docker.io
    => [proxy internal] load .dockerignore
    => transferring context: 2B
    => [proxy internal] load build context
    => transferring context: 14.15kB
    => [proxy 3/4] COPY ./proxy_server /app
    => [proxy 4/4] RUN pip install -r requirements.txt
    => [frontend internal] load .dockerignore
    => transferring context: 2B
    => [frontend 1/5] FROM docker.io/library/node:16@sha256:f77a1aef2da8d83e45ec990f45df50f1a286c5fe8bbfb8c6e4246c6389705cb
    => resolve docker.io/library/node:16@sha256:f77a1aef2da8d83e45ec990f45df50f1a286c5fe8bbfb8c6e4246c6389705cb
    => [frontend internal] load build context
    => transferring context: 3.43MB
    => CACHED [frontend 2/5] WORKDIR /app
    => CACHED [frontend 3/5] COPY package*.json ./
    => CACHED [frontend 4/5] RUN npm install
    => [frontend 5/5] COPY . .
    => [frontend] exporting to image
    => exporting layers
    => exporting manifest sha256:876ed05f1b64ea7c74ee2c67bba2e066e5819e997fb43487851c1aacf62a12b
    => exporting config sha256:b84f122ca7096706690baef4ab365f3d6fbvc1a2f77139a8b6140f5f3adea36
    => exporting attestation manifest sha256:1fd01c4029b5b599ca409fdb079425cfbe30830ff9d55762182c71fb86e7f763
    => exporting manifest list sha256:4130ee8abb4c9e9f8fcb67db585feaba079605aa56c8c98ee2ecc4a0ecf5404
    => naming to docker.io/library/hakaton-frontend:latest
    => unpacking to docker.io/library/hakaton-frontend:latest
    => [proxy] exporting to image
    => exporting layers
    => exporting manifest sha256:a217f67ec743766ef1940f72d021cbd82f6f19887da7caef8433fda5a067ec760
    => exporting config sha256:ad7d2d528f24815f218ba06442f2d3800c3844f80387b002bae0e7c1248ee73f

→ docker-compose up
time="2025-03-02T19:13:58+03:00" level=warning msg="D:\BSUIR\4th_Semester\IGI\353502_KOZELLO_13\IGI\LR2\HAKATON\docker-compose.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 4/4
  ✓ Container redis_cache      Created
  ✓ Container django_backend   Recreated
  ✓ Container proxy_server     Recreated
  ✓ Container react_app        Recreated
Attaching to django_backend, proxy_server, react_app, redis_cache
redis_cache | 1:C 02 Mar 2025 16:14:07.933 * oO0oO0oOoO0o Redis is starting oO0oO0oOoO0o
redis_cache | 1:C 02 Mar 2025 16:14:07.933 * Redis version=7.4.2, bits=64, commit=00000000, modified=0, pid=1, just started
redis_cache | 1:C 02 Mar 2025 16:14:07.933 # Warning: no config file specified, using the default config. In order to specify a config file use redis-server /path/to/redis.conf
redis_cache | 1:M 02 Mar 2025 16:14:07.933 * monotonic clock: POSIX clock_gettime
redis_cache | 1:M 02 Mar 2025 16:14:07.937 * Running mode=standalone, port=6379.
redis_cache | 1:M 02 Mar 2025 16:14:07.938 * Server initialized
redis_cache | 1:M 02 Mar 2025 16:14:07.939 * Loading RDB produced by version 7.4.2
redis_cache | 1:M 02 Mar 2025 16:14:07.939 * RDB age 18094 seconds
redis_cache | 1:M 02 Mar 2025 16:14:07.939 * RDB memory usage when created 1.02 Mb
redis_cache | 1:M 02 Mar 2025 16:14:07.939 * Done loading RDB, keys loaded: 0, keys expired: 0.
redis_cache | 1:M 02 Mar 2025 16:14:07.939 * DB loaded from disk: 0.001 seconds
redis_cache | 1:M 02 Mar 2025 16:14:07.939 * Ready to accept connections tcp
proxy_server | * Serving Flask app 'proxy'

```

☐	hakatон	-	-	-	0%	22 seconds ago			
☐	redis_cache	b4960a52c07e	redis:alpine	6379:6379	0%	23 seconds ago			
☐	django_backend	3e7130710d0b	hakaton-django:<none>	8000:8000	0%	23 seconds ago			
☐	proxy_server	6ddc6fc01664	hakaton-proxy:<none>	8080:8080	0%	22 seconds ago			
☐	react_app	06dc4ad6e416	hakaton-frontend:<none>	3000:3000	0%	22 seconds ago			

FCADHACK 2024

На главную
AdminPanel
Выйти

Фильтры
Применить параметры

Контактные Данные

Email
Endpoint
Номер телефона
Логин
Пароль
Timestamp
Message

Полученные данные
Очистить данные
Загрузить данные
Отключиться

email	name	surname	patronymic	birth date	gender	phone number	age	user id	login
*	—	—	—	—	—	—	—	892763239	alla4329
*	—	—	—	—	—	*	—	7824678117	vasiliy1986
*	—	—	—	—	—	*	—	8528117957	afanasiy57
*	—	—	—	—	—	—	—	6699331737	ignat2515
*	—	—	—	—	—	—	—	6974837078	olga1967
*	—	—	Яковна	01.10.1971	Женщина	—	—	342149180	oksana.bat
*	—	—	—	—	—	*	—	1983989279	pavel1973
*	—	—	—	—	—	—	—	1927690800	rada9004

5. Настроить сети и тома для обеспечения связи между контейнерами и сохранения данных (исходные данные, логин, пароль и т.д.)

Выполнено в предыдущем пункте

6. Разместите результат в созданный репозиторий в DockerHub

```

→ docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
mdcxxi/django-backend latest     6184361748b7  3 minutes ago  771MB
mdcxxi/proxy-server  latest     66153541f016  31 minutes ago 401MB
mdcxxi/react-app     latest     ea4fad16ab82  32 minutes ago 3.74GB
geometric_lib        1.0        ebad179d731b  About an hour ago 1.47GB
python               latest     385ccb8304f6  3 weeks ago   1.47GB
redis                alpine     02419de7eddf  7 weeks ago   60.6MB
docker/getting-started latest     d79336f4812b  2 years ago   73.9MB
→ docker push mdcxxi/django-backend:latest
The push refers to repository [docker.io/mdcxxi/django-backend]
b9a201c1d152: Pushed
7446f41c9665: Pushed
5dbb3b698b72: Pushed
0c5ce2cb4ecc: Pushed
8356458b3f2d: Pushed
7cf63256a31a: Pushed
d27b70128a63: Pushed
183f0922284a: Pushed
a094a41307c3: Pushed
latest: digest: sha256:6184361748b74ba2277ac93a067b6d6c8839549ff98aeb6278d5306b5a9a1c3b size: 856

```

```

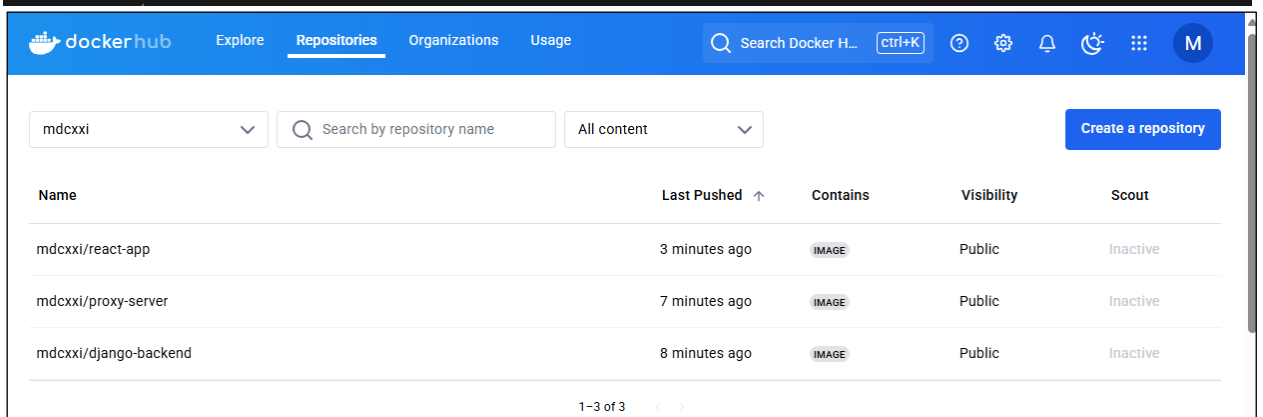
→docker push mdcxxi/proxy-server:latest
The push refers to repository [docker.io/mdcxxi/proxy-server]
ed087bc81923: Pushed
5dbb3b698b72: Mounted from mdcxxi/django-backend
7cf63256a31a: Mounted from mdcxxi/django-backend
d27b70128a63: Mounted from mdcxxi/django-backend
0c5ce2cb4ecc: Mounted from mdcxxi/django-backend
31a2fd5123d6: Pushed
183f0922284a: Mounted from mdcxxi/django-backend
641cfc5b7757: Pushed
latest: digest: sha256:66153541f016b69356e5f809306ec1155d511cc58dee43cad0e92877aabf9962 size: 856

```

```

→docker push mdcxxi/react-app:latest
The push refers to repository [docker.io/mdcxxi/react-app]
67b7c7973b6f: Pushed
4e9095b5cfad: Pushed
ee7d78be1eb9: Pushed
513d77925604: Pushed
7e9bf114588c: Pushed
d66ddacf56ef: Pushed
ffd9397e94b7: Pushed
3fb7b7b1289a: Pushed
0e421f66aff4: Pushed
311da6c465ea: Pushed
ca266fd61921: Pushed
ee0b97d47c25: Pushed
ae3b95bbaa61: Pushed
latest: digest: sha256:ea4fad16ab82884e020941802089220e7fe9ee129cdaa9e221e4d4de7a204634 size: 856

```



The screenshot shows the Docker Hub interface for the user 'mdcxxi'. The 'Repositories' tab is active, displaying a list of three repositories: 'mdcxxi/react-app', 'mdcxxi/proxy-server', and 'mdcxxi/django-backend'. Each repository entry shows its last push time (3, 7, and 8 minutes ago respectively), a status of 'IMAGE', and 'Public' visibility. A 'Scout' button is present for each repository, all of which are currently 'Inactive'.

Name	Last Pushed	Contains	Visibility	Scout
mdcxxi/react-app	3 minutes ago	IMAGE	Public	Inactive
mdcxxi/proxy-server	7 minutes ago	IMAGE	Public	Inactive
mdcxxi/django-backend	8 minutes ago	IMAGE	Public	Inactive

7. Выполните следующие действия с целью изучить особенности сетевого взаимодействия:

- Получить информацию о всех сетях, работающих на текущем хосте и подробности о каждом типе сети

```

→docker network ls
NETWORK ID      NAME                DRIVER             SCOPE
da8a4a0c23ef    bridge             bridge             local
15fdd77b8a4c    hakaton_app_network bridge             local
fd49f7dd03e5    host               host               local
f0cd8f697d9d    none              null               local

```



```
→ docker network inspect bridge
```

```
[
  {
    "Name": "bridge",
    "Id": "da8a4a0c23ef1d6e7be5182e3df8bfa09a81fc0e0c34da7cce3b5f3025c0242c",
    "Created": "2025-03-02T14:27:40.482327028Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {
      "com.docker.network.bridge.default_bridge": "true",
      "com.docker.network.bridge.enable_icc": "true",
      "com.docker.network.bridge.enable_ip_masquerade": "true",
      "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
      "com.docker.network.bridge.name": "docker0",
      "com.docker.network.driver.mtu": "1500"
    },
    "Labels": {}
  }
]
```

```
→ LR2 ↻ (🔗 igi_lr2)
```

```
→ docker network inspect hakaton_app_network
```

```
[
  {
    "Name": "hakaton_app_network",
    "Id": "15fdd77b8a4c4b86fb783d4f77af995b2ac8ae528e6ccf93d64e0e79a9f0c7dd",
    "Created": "2025-03-02T11:33:19.553311875Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
```

```

        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {},
        "Labels": {
            "com.docker.compose.network": "app_network",
            "com.docker.compose.project": "hakaton",
            "com.docker.compose.version": "2.29.2"
        }
    }
}
]
→ LR2 ⚡ (❗ igi_lr2)
→ docker network inspect host
[
    {
        "Name": "host",
        "Id": "fd49f7dd03e55158cf1a40a5f5293ea168fd32eb2b320df267704ffe9a815294",
        "Created": "2024-10-13T16:13:59.649545981Z",
        "Scope": "local",
        "Driver": "host",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": null
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {},
        "Labels": {}
    }
]

```

```

→ docker network inspect none
[
    {
        "Name": "none",
        "Id": "f0cd8f697d9d1748d48eb0c7764d41c12dc99b6c47bbebc32bf27131331113c7",
        "Created": "2024-10-13T16:13:59.631118203Z",
        "Scope": "local",
        "Driver": "null",
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": null
        },
    }
]

```

```

    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]

```

- Создать свою собственную сеть bridge, проверить, создана ли она, запустить Docker-контейнер в созданной сети, вывести о ней всю информацию(включая IP-адрес контейнера), отключить сеть от контейнера

```

→ docker network create --driver bridge bridge-net
f7628cfda78af316a207b8daf38501a60ff9ce9d330b3f0f3d7932e3f5a5af62
→ docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
da8a4a0c23ef	bridge	bridge	local
f7628cfda78a	bridge-net	bridge	local
15fdd77b8a4c	hakaton_app_network	bridge	local
fd49f7dd03e5	host	host	local
f0cd8f697d9d	none	null	local

```

→ docker run -d --network bridge-net --name someapp nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
103f50cb3e9f: Download complete
9dd21ad5a4a6: Download complete
943ea0f0c2e4: Download complete
d014f92d532d: Download complete
bf9acace214a: Download complete
513c3649bb14: Download complete
Digest: sha256:9d6b58feebd2dbd3c56ab585333d627cc6e281011cfd6050fa4bcf2072c9496
Status: Downloaded newer image for nginx:latest
4238614cea5ab79a540287c862a0ae8d56d70bdc4461c7f0b3447654a7d12baf
→ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
4238614cea5a	nginx	"/docker-entrypoint..."	15 seconds ago	Up 14 seconds	80/tcp	someapp

```

→ docker network inspect bridge-net
[
  {
    "Name": "bridge-net",
    "Id": "f7628cfda78af316a207b8daf38501a60ff9ce9d330b3f0f3d7932e3f5a5af62",
    "Created": "2025-03-02T17:52:24.163367539Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,

```

```

    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "4238614cea5ab79a540287c862a0ae8d56d70bdc4461c7f0b3447654a7d12baf": {
        "Name": "someapp",
        "EndpointID": "707805520d7e6a3f214e92bd9fe60e00edd08a69d311b48489f7796bc007d88e",
        "MacAddress": "02:42:ac:13:00:02",
        "IPv4Address": "172.19.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {},
    "Labels": {}
  }
]

→ docker network disconnect bridge-net someapp
→ docker network inspect bridge-net
[
  {
    "Name": "bridge-net",
    "Id": "f7628cfda78af316a207b8daf38501a60ff9ce9d330b3f0f3d7932e3f5a5af62",
    "Created": "2025-03-02T17:52:24.163367539Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]

```

- Создать еще одну сеть bridge, вывести о ней всю информацию, запустить в ней три контейнера, подключиться к любому из контейнеров и пропинговать два других из оболочки контейнера, убедиться, что между контейнерами происходит общение по IP-адресу

```

→ LR2 ⚡ (❯ igi_lr2)
→ docker network create myNet
5ed5d61d5f8760ddfd3a1d0475a80e6bc9f97bbc0cc8c7cfd02389ba228b37ef
→ docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
da8a4a0c23ef	bridge	bridge	local
f7628cfda78a	bridge-net	bridge	local
15fdd77b8a4c	hakaton_app_network	bridge	local
fd49f7dd03e5	host	host	local
5ed5d61d5f87	myNet	bridge	local
f0cd8f697d9d	none	null	local

```

→ docker run -it -d --name A1 --network myNet alpine ash
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
Digest: sha256:a8560b36e8b8210634f77d9f7f9efd7ffa463e380b75e2e74aff4511df3ef88c
Status: Downloaded newer image for alpine:latest
6fc1359ef28d049451ce3794b0e94fe115f4e6d1a2bf5166e4bbc68f096237ed
→ docker run -it -d --name A2 --network myNet alpine ash
20f28d0becb9f4b66ede0234a695c11fc5bc310f2e3e2a9fbff0d0d619fc5ccb
→ docker run -it -d --name A3 --network myNet alpine ash
96ce90fab872e34c5a4f55d672afc2cb154d02169858d5e2a9b9b9fba906d24a
→ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
96ce90fab872	alpine	"ash"	13 seconds ago	Up 13 seconds		A3
20f28d0becb9	alpine	"ash"	18 seconds ago	Up 18 seconds		A2
6fc1359ef28d	alpine	"ash"	28 seconds ago	Up 28 seconds		A1

```

→ LR2 ⚡ (❯ igi_lr2)

```

```

→ docker network inspect myNet
[
  {
    "Name": "myNet",
    "Id": "5ed5d61d5f8760ddfd3a1d0475a80e6bc9f97bbc0cc8c7cfd02389ba228b37ef",
    "Created": "2025-03-02T19:00:41.364054684Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,

```

```

"Containers": {
  "20f28d0becb9f4b66ede0234a695c11fc5bc310f2e3e2a9fbff0d619fc5ccb": {
    "Name": "A2",
    "EndpointID": "3aa57c748b9bd6b450b3e0bb7b7159d82fca1063606f1b14b79567b321c0742b",
    "MacAddress": "02:42:ac:14:00:03",
    "IPv4Address": "172.20.0.3/16",
    "IPv6Address": ""
  },
  "6fc1359ef28d049451ce3794b0e94fe115f4e6d1a2bf5166e4bbc68f096237ed": {
    "Name": "A1",
    "EndpointID": "f9469dd6ff86ab1c83db244da1d3a6afe70b03f71e6f3ad1e115f5e28eef1555",
    "MacAddress": "02:42:ac:14:00:02",
    "IPv4Address": "172.20.0.2/16",
    "IPv6Address": ""
  },
  "96ce90fab872e34c5a4f55d672afc2cb154d02169858d5e2a9b9b9fba906d24a": {
    "Name": "A3",
    "EndpointID": "520299e117ab7319dad3ede7112ad2ee30a4ef680441f1ebade2bec8682b2ade",
    "MacAddress": "02:42:ac:14:00:04",
    "IPv4Address": "172.20.0.4/16",
    "IPv6Address": ""
  }
},
"Options": {},
"Labels": {}
}

```

```

→ docker container attach A2
/ # ping -c 2 A1
PING A1 (172.20.0.2): 56 data bytes
64 bytes from 172.20.0.2: seq=0 ttl=64 time=1.249 ms
64 bytes from 172.20.0.2: seq=1 ttl=64 time=0.152 ms

--- A1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.152/0.700/1.249 ms
/ # ping -c 2 A3
PING A3 (172.20.0.4): 56 data bytes
64 bytes from 172.20.0.4: seq=0 ttl=64 time=2.475 ms
64 bytes from 172.20.0.4: seq=1 ttl=64 time=0.229 ms

--- A3 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.229/1.352/2.475 ms

```

- Создать свою собственную сеть overlay, проверить, создана ли она, вывести о ней всю информацию

```

→ docker network create --driver overlay myOverlay
Error response from daemon: This node is not a swarm manager. Use "docker swarm init" or "docker swarm join" to connect this node to
swarm and try again.
→ docker swarm init
Swarm initialized: current node (vurfwk7qjt3m3doyo2bifufoenf) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-lgsohie8p9jeq93qkm8qdvace44vu3o79v5mg5f3y1cmlllyae-cd89110hn41yh60h52341rhnv 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.

→ docker network create --driver overlay myOverlay
xlxam00t1hl7t4iifbbhe7zta
→ docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
da8a4a0c23ef	bridge	bridge	local
f7628cfda78a	bridge-net	bridge	local
8a23fb246aa0	docker_gwbridge	bridge	local
15fdd77b8a4c	hakaton_app_network	bridge	local
fd49f7dd03e5	host	host	local
vef4vs8k8bm5	ingress	overlay	swarm
5ed5d61d5f87	myNet	bridge	local
xlxam00t1hl7	myOverlay	overlay	swarm
f0cd8f697d9d	none	null	local

```

→ LR2 ↵ (v igi_lr2)

```

2 hours ago

```

→ docker network inspect myOverlay
[
  {
    "Name": "myOverlay",
    "Id": "xlxam00t1hl7t4iifbbhe7zta",
    "Created": "2025-03-02T19:13:07.915741136Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.1.0/24",
          "Gateway": "10.0.1.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": null,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4097"
    },
    "Labels": null
  }
]

```

- Создать еще одну сеть overlay, проверить, создана ли она, вывести о ней всю информацию, удалить сеть

```

→ docker network create --driver overlay newOverlay
xb8lhm547djynbtrcoabqelka
→ docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
da8a4a0c23ef	bridge	bridge	local
f7628cfda78a	bridge-net	bridge	local
8a23fb246aa0	docker_gwbridge	bridge	local
15fdd77b8a4c	hakaton_app_network	bridge	local
fd49f7dd03e5	host	host	local
vef4vs8k8bm5	ingress	overlay	swarm
5ed5d61d5f87	myNet	bridge	local
xlxam00t1hl7	myOverlay	overlay	swarm
xb8lhm547dgy	newOverlay	overlay	swarm
f0cd8f697d9d	none	null	local

```

→ docker network inspect newOverlay
[
  {
    "Name": "newOverlay",
    "Id": "xb8lhm547djynbtrcoabqelka",
    "Created": "2025-03-02T19:16:02.174441761Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.2.0/24",
          "Gateway": "10.0.2.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": null,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4098"
    },
    "Labels": null
  }
]

```

```
→ docker network rm newOverlay
```

```
newOverlay
```

```
→ docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
da8a4a0c23ef	bridge	bridge	local
f7628cfda78a	bridge-net	bridge	local
8a23fb246aa0	docker_gwbridge	bridge	local
15fdd77b8a4c	hakaton_app_network	bridge	local
fd49f7dd03e5	host	host	local
vef4vs8k8bm5	ingress	overlay	swarm
5ed5d61d5f87	myNet	bridge	local
xlxam00t1hl7	myOverlay	overlay	swarm
f0cd8f697d9d	none	null	local

```
→ LR2 ↩ (P iqi lr2)
```

- Попробовать создать сеть host, сохранить результат в отчет.

```
→ docker network create --driver host myHost
```

```
Error response from daemon: only one instance of "host" network is allowed
```